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Roll No

EX-7004(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Reliability Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define Reliability. 4
b) Explain with example Probability distribution function. 5
c) Justify the use of exponential distribution in reliability of non repairable systems. 5
2. a) Derive the expression for Reliability interns of hazard rate. 7
b) Explain in detail the different techniques employed in improving the reliability. 7
3. a) Derive the expression Reliability evolution of series and parallel systems. 7
b) Explain the conditional probability method for Reliability evolution for non series-Parallel System. 7
4. a) Explain Markov process for Reliability evolution of repairable systems. 7
b) What is failure rate and how it is used in the calculation of Reliability? 7

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- 5. a) Explain about the Reliability Testing. 7
- b) Explain about the parametric methods. 7
- 6. a) Explain about the state enumeration method for evaluating failure frequency. 7
- b) How absorbing state are useful in reliability evolution? 7
- 7. a) How reliability and quality are associated? Discuss. 7
- b) What do you mean by Tie set and Cut set? 7
- 8. Write short notes on 14
- a) Frequency balance approach
- b) Explain Mean Time To Failure (MTTF).
